

Practice 1: Likelihood language

Name: _____ Date: _____

Use impossible, unlikely, equally likely, likely, or certain. Explain your choice.

1. A standard six-sided die is rolled. The result is a number from 1 through 6.

Work space

2. A bag has 9 blue tiles and 1 red tile. Which color is more likely? How do you know?

Work space

Practice 2: Sample spaces

List every possible outcome, with no repeats.

1. A coin is tossed once.

Work space

2. A spinner has equal sections labeled 1, 2, 3, and 4. It is spun once.

Work space

Practice 3: Favorable outcomes

Find the number of total outcomes and favorable outcomes.

1. A die is rolled. What is the probability of rolling a 5?

Work space

2. A die is rolled. What is the probability of rolling an even number?

Work space

Practice 4: Probability as a fraction

Write each probability as a fraction in simplest form when possible.

1. A jar has 4 green, 3 yellow, and 1 purple marble. What is $P(\text{yellow})$?

Work space

2. From the same jar, what is $P(\text{not purple})$?

Work space

Practice 5: Impossible and certain events

1. A bag contains only red and white counters. What is the probability of drawing a blue counter?

Work space

2. A box contains 12 pencils, all sharpened. What is the probability of choosing a sharpened pencil?

Work space

Practice 6: Complementary events

Use $P(\text{event}) + P(\text{not event}) = 1$.

1. A spinner has 8 equal sections; 3 are orange. Find $P(\text{not orange})$.

Work space

2. The probability that a bus is early is $\frac{1}{5}$. Find the probability that it is not early.

Work space

Practice 7: Comparing probabilities

1. Bag A has 2 red tiles and 6 blue tiles. Bag B has 4 red tiles and 8 blue tiles. Which bag gives a greater chance of red? Show work.

Work space

2. Compare $\frac{3}{10}$ and $\frac{1}{4}$. Which event is more likely? Explain.

Work space

Practice 8: Fairness

1. A game uses a fair coin. Is calling heads a fair way to choose between two players? Explain.

Work space

2. A spinner has 3 equal red sections and 1 equal blue section. Is it fair for red and blue players? Why or why not?

Work space

Practice 9: Theoretical probability

1. A fair die is rolled 60 times. About how many 6s would you expect? Explain.

Work space

2. A bag has 5 green and 5 yellow counters. In 40 draws with replacement, about how many green draws are expected?

Work space

Practice 10: Experimental probability

Experimental probability is favorable results divided by trials.

1. A coin lands heads 17 times in 30 tosses. Find the experimental probability of heads.

Work space

2. A spinner lands on blue 9 times in 25 spins. Find the experimental probability of blue.

Work space

Practice 11: Predicting from data

1. A die is rolled 120 times. A 4 appears 18 times. Use the data to predict how many 4s may appear in 600 rolls.

Work space

2. A red result occurred 12 times in 20 trials. Predict red results in 100 similar trials.

Work space

Practice 12: Tables and probability

Use the table to answer each question.

Color	Red	Blue	Green	Yellow
Count	6	5	3	2

1. How many tiles are there? Find $P(\text{green})$.

Work space

2. Find $P(\text{red or blue})$.

Work space

Practice 13: Two-step outcomes

For two independent coin tosses, list ordered outcomes.

1. List the sample space for two tosses and find $P(\text{two heads})$.

Work space

2. Find $P(\text{exactly one head})$ for two tosses.

Work space

Practice 14: Two-color draws

Choose one counter, replace it, then choose again.

1. A bag has 2 red and 1 blue counter. List the four possible color pairs and find $P(\text{red then red})$.

Work space

2. For the same bag and replacement, find $P(\text{one red and one blue in either order})$.

Work space

Practice 15: Reading claims

1. Sam says, “A probability of $\frac{7}{4}$ is possible.” Is Sam correct? Explain using the probability range.

Work space

2. Lee says, “If an event has probability $\frac{1}{2}$, it must happen exactly half the time.” Evaluate the claim.

Work space

Practice 16: Showing reasoning

Write a clear conclusion with an equation or fraction.

1. A box has 5 apples and 7 oranges. What is the probability of an apple? Show numerator and denominator.

Work space

2. Explain why a probability of 0 means impossible and 1 means certain. Give one example of each.

Work space

Practice 17: Word problems

1. A prize wheel has 10 equal spaces: 4 toys, 3 books, 2 games, and 1 sticker. What is $P(\text{a book or game})$?

Work space

2. In a class survey, 15 of 24 students prefer apples. What fraction prefer another fruit? Simplify.

Work space

Practice 18: Design a probability model

1. Design a fair spinner for 3 players. Describe the equal sections and justify fairness.

Work space

2. Design a spinner where blue has probability $\frac{1}{4}$. Describe one possible spinner.

Work space

Practice 19: Mixed review

1. A bag has 3 red, 4 blue, and 5 green cubes. Find $P(\text{not green})$ and explain your count.

Work space

2. A fair die is rolled 90 times. Predict the number of odd results and state the assumption used.

Work space

Practice 20: Explain and justify

1. Which is more likely: rolling a number greater than 4 on a die, or drawing a red tile from a bag with 3 red and 7 blue tiles? Compare the fractions.

Work space

2. Write a short rule for finding theoretical probability, then apply it to $P(\text{rolling less than 3})$ on a fair die.

Work space